# Narayana Karthik Kumar

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 7.42 |
| Sri Chaitanya CBSC & PU College | Class XII (CBSE) | 2023 | 89.20% |
| Sri Chaitanya Techno School | Class X (CBSE) | 2021 | 82.00% |

## SCHOLASTIC ACHIEVEMENTS

**Ascent Quest Coding Contest - Certificate of Merit | The Ascent Circle, HackerRank**

- Achieved a perfect score of 100/100 in the Ascent Quest Coding Contest (Season 1), demonstrating problem-solving and coding skills.
- Awarded by The Ascent Circle.

## PROJECTS

**Virtual Memory Management and Copy-on-Write Fork in gemOS**

Technologies: C, gemOS Kernel, Operating Systems

- Implemented mmap(), munmap(), and mprotect() system calls in a Linux-like kernel.
- Supported dynamic Virtual Memory Area (VMA) creation, protection flags, and VMA merging and splitting.
- Developed a custom page-fault handler supporting lazy allocation and four-level page-table walking.
- Designed and implemented cfork() with Copy-on-Write semantics using shared page tables, reference-counted physical frames, and address-space tracking.
- Validated correctness and edge-case behaviour through user-level testing in init.c.

**Multithreaded Sudoku Validator -- Static and Dynamic Scheduling**

Technologies: C, C++, POSIX Threads, Atomics

- Engineered a multithreaded validator for NxN Sudoku grids, concurrently validating rows, columns, and subgrids.
- Implemented chunk and mixed scheduling in C and dynamic work allocation in C++ using a shared counter and configurable task increment.
- Integrated and evaluated multiple mutual-exclusion strategies using C++ atomics to synchronize access to shared state.
- Conducted experiments across Sudoku sizes, thread counts, and task granularities to assess scalability and synchronization overhead.
- Recorded thread actions, critical-section entry and exit, task allocation, and validation results using timestamps.

**Bounded Buffer Producer-Consumer**

Technologies: C++, Semaphores, Locks, Multithreading

- Implemented the producer-consumer problem using semaphores and locks for synchronized access to a shared circular buffer.
- Simulated thread workloads using exponential delays and logged buffer operations with timestamps.
- Benchmarked performance across varying delay and thread-count ratios.
- Visualized comparative performance and analyzed synchronization overhead and bottlenecks in lock-based and semaphore-based approaches.

**Stability Analysis of Fractional-Order Systems Using Machine Learning**

- Designed a hybrid framework combining fractional calculus and machine learning to model and analyze dynamical systems with memory and non-local behaviour.
- Explored ML-based approximations of system solutions and identification of stability regions.
- Studied the application of ML to complex systems where traditional analytical approaches are challenging.

**Transcriptomics -- RNA-seq Differential Expression Analysis**

Technologies: Python, Linux, MySQL, DESeq2, TCGA/GEO

- Developed a Linux-based RNA-seq analysis pipeline using public TCGA/GEO datasets to study differential gene expression between cancer and normal tissue samples.
- Performed data preprocessing, normalization, and statistical testing using DESeq2 workflows.
- Designed and managed a MySQL database for gene expression counts, sample metadata, and intermediate results.
- Generated volcano plots and clustered heatmaps of differentially expressed genes using Python.
- Interpreted biological significance using fold-change thresholds and statistical significance.

**Predictive Modeling of Startup Competitiveness -- Indian Tech Ecosystem** *(Academic Research Project, Ongoing)*

- Extended a structural m-TISM/MICMAC framework of nine competitiveness factors into an ML-based predictive pipeline, using PCA-derived and clustering-based features to model startup competitiveness/success outcomes
- Applied feature-importance analysis to empirically validate expert derived factor hierarchies against data-driven predictors
- Combined qualitative case study insights (Indian unicorns/soonicorns) with quantitative modeling to bridge theoretical and empirical approaches to competitiveness research

## WORK EXPERIENCE

**Applied AI Intern -- Prof. Praveen Tammana, IIT Hyderabad**

- Developed a Hackathon Evaluation and Verification Engine combining static code analysis, complexity and testing checks, and problem-statement coverage mapping to produce evidence-backed technical evaluations.
- Built a sandboxed dynamic-validation pipeline to safely execute untrusted submissions using isolated, resource-limited containers.
- Exercised API endpoints, verified data flow, and detected hardcoded or non-functional implementations.
- Integrated Gemini-based evaluation to generate structured technical assessments with supporting evidence, identify critical red flags, and formulate targeted probing questions for jury review.
- Built an AI Research Navigator over a curated corpus of 50 documents spanning LLMs, RAG, agents, alignment, and evaluation.
- Used normalized metadata and topic tags to organize technical research resources.

**Drone Firmware Developer -- Amani Automations**

- Designed and implemented custom flight-control firmware using C++ on STM32 microcontrollers.
- Developed PID-based attitude stabilization, SBUS signal decoding, and sensor data fusion involving IMU, barometer, and magnetometer.
- Implemented motor control using PWM and ESCs, along with failsafe mechanisms.
- Contributed to the design, programming, and testing of a custom flight controller.
- Successfully achieved stable drone flight using the custom-designed and programmed flight controller.

## SKILLS

**Engineering and Research Tools:** NumPy, Matplotlib, Jupyter, Quantum ESPRESSO, VESTA, Vivado, SolidEdge, SolidWorks, LaTeX.

**Embedded Systems and Hardware:** STM32 HAL, PlatformIO, Arduino, Raspberry Pi, ESP32, Sensor integration, PID control, PWM/ESC motor control.

**Machine Learning and AI:** PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, LLMs, Retrieval-Augmented Generation (RAG), AI agents, LangChain, prompt engineering, Gemini API, Anthropic API.

**Systems and Software:** Linux, Ubuntu, Git, GitHub, POSIX threads, mutexes, semaphores, C++ atomics, MySQL, Operating systems, data structures, algorithms, concurrency, virtual memory, API testing.

**Programming Languages**: C, C++, Python, Julia, HTML, CSS, JavaScript, MATLAB, Verilog

**Embedded Systems and Hardware:** STM32 HAL, PlatformIO, Arduino, Raspberry Pi, ESP32, Sensor integration, PID control, PWM/ESC motor control.

## RELEVANT COURSES

**Mathematics**: Calculus, Probability, Statistics, Linear Algebra, Transformation Techniques, Differential Equations, Linear Algebra, Complex Variables

**Computer Science**: Introduction to Programming, Software Development Fundamentals, Data Structures and Algorithms, Operating Systems, Database Management Systems, Fundamentals of Scientific Computing, Computer-Aided Numerical Methods, Computational Methods in Material Science

**Basic Engineering:** *Introduction to Programming, Basic Electrical Engineering, Modern Physics, Digital Fabrication*

**High Performance Computating:** Distributed Computing, Parallel and Concurrent Programming, Mesoscale Micro Structure Modeling.

## POSITIONS OF RESPONSIBILITY

**Class Representative:**

Represented the academic and welfare interests of the class to faculty and administration. Actively participated in curriculum feedback, policy discussions, and conflict resolution.

**Shuffle Management Head :**

Led the planning and coordination of Shuffle, a cultural club at IIT Hyderabad. Oversaw event scheduling, team logistics to ensure seamless execution for Inter IIT Cultural 7.0

**Digital Fabrication Teaching Assistant:**

Assisted in the Digital Fabrication course, guiding students through CAD modelling and 3D printing. Helped in lab sessions and provided technical mentorship.

**Departmental Representative of Inter-Departmental Sports & Cultural Fest:**

Led department participation in inter-departmental fests; coordinated events, team logistics, and performance tracking across sports and cultural domains.

**Office of Career Services -- Placement Coordinator:**

- Coordinated company outreach and communication to onboard recruiters for internship opportunities across engineering branches.
- Supported internship-drive scheduling, assessment rounds, and coordination between candidates and company representatives.
- Facilitated company assessments, including test administration and candidate logistics.

**Office of Career Services Internship Coordinator:**

Actively pitching companies to onboard for internship opportunities across various engineering branches. Played a key role in the internship drive coordination, ensuring smooth communication, scheduling, and logistics. Took charge of conducting assessment rounds for allocated companies, managing everything from test facilitation to coordination with student candidates and company representatives.

**Finance Head -- EBSB:** Managed financial coordination and budgeting responsibilities for EBSB activities.

**Deputy Contingent Leader -- Inter-IIT Cultural Meet 8.0**

- Co-led IIT Hyderabad's contingent for Inter-IIT Cultural Meet 8.0.
- Coordinated team mobilization, practice schedules, and on-ground event logistics.

**Overall Coordinator -- Otaku Club**

- Headed club activities, including anime screenings, fandom-based events, and community engagement.
- Coordinated activities to develop the Japanese-media community at IIT Hyderabad.

**Multimedia Head -- Media Council, IIT Hyderabad:** Led the multimedia team in content design, video editing, and promotional campaigns. Directed creative projects for Gymkhana and institute events.

## CERTIFICATIONS

**Python - Kaggle | Kaggle Learn:** Successfully completed the Python course on Kaggle Learn.

## EXTRACURRICULAR

Institute Hockey League winner for three years.

Participated in Inter-IIT Cultural Meet 7.0 as part of the Dance Club.

Member of the Inter-IIT Cultural Meet 7.0 dance crew and Photoshop Battle team.

Member of the Inter-IIT Cultural Meet 8.0 dance crew and Photoshop Battle team.

Hockey NSO Team Member, IIT Hyderabad (2023-Present).